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1.0 INTRODUCTION

1.1 Purpose of this document

The purpose of this document is to describe the requirement and guidance of electrical and instrument (E&I) tie in document. E&I tie in document is defined that drawing and list which indicate any information related to determine the HHI yard construction interface. This document states not only what kinds of data to be specified but also how to describe the required data. Tie in document will be dominated document to determine yard cable list, yard cable entry sizes, yard cable termination, package tie in location and others.

2.0 DEFINITIONS

- COMPANY: CLIENT OF PROJECT
- CONTRACTOR(HHI): HYUNDAI HEAVY INDUSTRIES CO. LTD
- VENDOR: PACKAGE supplier /manufacturer
- Yard Cables: Cables installed by CONTRACTOR at yard. It includes the loosed cables supplied by VENDOR.
- VENDOR cables: Cables installed by VENDOR.
- Special Cables: Cable type which is not specified on project cable specification. (To be supplied by VENDOR)
- Yard termination: Cable termination to be conducted by CONTRACTOR
- VENDOR termination: Cable termination to be conducted by VENDOR

3.0 GENERAL

E&I tie in document is required to indicate only data related to yard cables. Therefore, VENDOR shall ensure to exclude the VENDOR cables (installed by VENDOR) and VENDOR terminations (conducted by VENDOR) from this document and they should be shown in other relevant documents.

E&I tie in document shall include following information but not limited to:

A. The information of cable tray (tray location, size and others) for yard cables

B. Yard termination data

C. Yard cable information

D. Yard cable entry sizes / Cable gland sizes for equipment.

E. Ship loose E&I devices installation detail

F. IO data, load data.

G. Skid edge tie in location

CONTRACTOR will finalize the interface design such as cable tray tie in locations, terminations, cables and support/seat of ship loose E&I equipment based on this document and if any conflicts between other vendor documents, E&I tie in document will be priority to determine the E&I interface between VENDOR and CONTRACTOR. Therefore, if any construction and design impacts caused by the discrepancy between tie in document and other vendor data/actual equipment will be the responsibility of the VENDOR.

3.1 List & Priority of E&I Tie in document

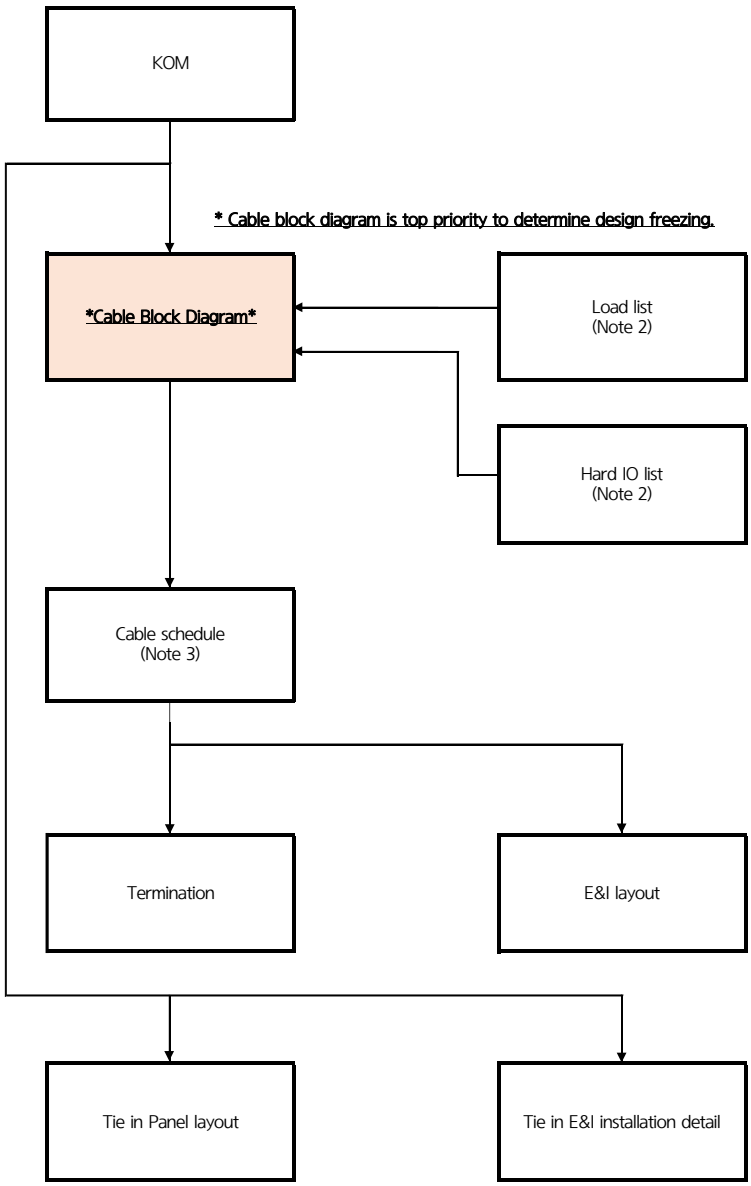
E&I tie in document shall consist of the following;

- Tie in cable block diagram/single line diagram
- Tie in cable schedule
- Tie in termination drawing
- Tie in E&I layout
- Tie in panel/cabinet layout
- Tie in E&I installation detail for ship loose devices.

Among above documents, cable block diagram is top priority to determine the cable number, cable type, E&I equipment tag number and others when there is any discrepancy among documents. Therefore, cable schedule, termination and E&I layout shall be submitted after incorporating the first comment on tie in cable block diagram in order to minimize the document cycling.

It is VENDOR's sole responsibility to incorporate the all data/comment on cable block diagram to other relevant document such as tie in cable schedule, tie in termination, tie in E&I layout and others.

3.2 Work flow chart for E&I Tie in document



A	CONTRACTOR provides with following
1	Cable tag numbering proceudre
2	Instrument numerbing, Instrument IO numerbing proceudre
3	Terminal block numbering proceudre
4	Wire marker numbering proceudre
5	Junction box, Control panel numbering proceudre
6	Elec. Equipment numbering proceudre
7	Project cable & gland selection table (Cable coding)
8	Cable tray segregation table

Note 1

If not availble at KOM, it will be supplied later

B	Determine following via cable block diagram.
1	Cable Tag number
2	E&I Equipment tag number
3	Cable type
4	Cable tray system
5	Required cable entry size for yard cable

Note 2

Data on tie in cable block diagram shall be in line with load list and Hard IO list.

C	Determine terminal block size/type for yard cable

Note 3

It's VENDOR's sole responsibility to incorporte data on cable block diagram to cable schedule.

D	Determine the CONTRACTOR termination
E	Determine the skid tie in location for tubing and cables

F	Determine yard cable tie in location for loosed equipment.
G	Determine seat design of cabinet / E&I equipment.

3.3 Document template and format requirement.

VENDOR is responsible for providing tie in document by using project tie in document template if applicable. When tie in document templates are not determined at bidding stage, they will be provided during detailed engineering stage (in Kick off meeting). Special attention shall be given to fill out all required data on tie in cable block diagram template/format according to project numbering requirement.

3.4 Project numbering requirement.

Below project numbering requirement will be provided during bidding stage or detailed engineering stage to VENDOR. VENDOR shall submit the tie in document after employing project numbering procedure.

- Cable tag numbering
- Instrument numbering
- IO numbering
- Terminal block numbering
- Wire marker numbering
- Junction box numbering
- Control panel numbering
- Electrical equipment numbering

3.5 Other document reference

CONTRACTOR will provide following project data during detailed engineering stage. VENDOR shall submit all tie in document by employing them.

- Project cable & gland selection table (Cable code)
- Cable tray segregation table

4.0 DOCUMENT GUIDANCE

This section gives the guidance for each tie in document. It is mandatory to comply following requirement when VENDOR submits E&I tie in document.

4.1 Tie in Cable block Diagram / Single line diagram

As mentioned in 3.1, it is master document to determine the relevant data of other tie in documents. Thus, VENDOR shall make sure to update other tie in documents according to information and comment on this document.

Tie in cable block diagram/single line diagram shall indicate all cables to be installed by CONTRACTOR. Specially, it should contain all network cables and earthing cables for control panels and equipment.

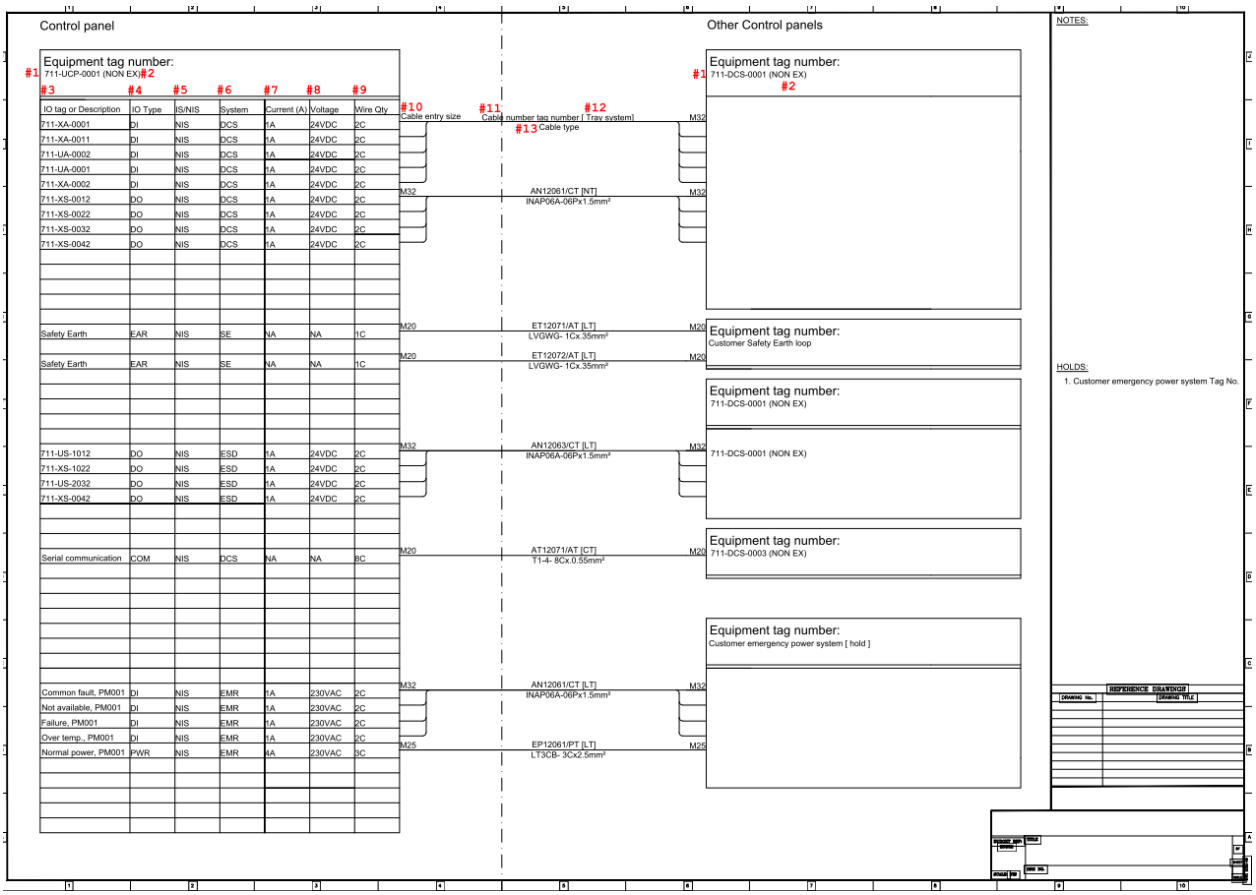
VENDODR shall submit this document by using the template / format supplied by CONTRACTOR. This template requests to fill out the following data and it is mandatory to fill out all required data as per project requirement/numbering procedure.

A. IO tag for instrument
B. Load/signal description for electric system
C. IO type
D. IS(Intrinsic safety) / NIS (Non Intrinsic safety)
E. System
F. Current (Power consumption)
G. Voltage (Power rating)
H. Wire Quantity
I. From / To equipment tag number/description
J. Cable entry size
K. Cable tag numbering
L. Cable type
M. Cable tray system

Following describes the detail guidance of cable block diagrams / single line diagrams in many cases.

Case 1: Panel (cabinet) to panels (cabinets)

- It shall be generated via project template for “panel to panel”.
- VENDOR should indicate only yard cables which is installed by CONTRACTOR.
- All data from # 1~#13 on Figure 1 shall be specified by VENDOR according following guidance which is general requirement for all cases.



- **Figure 1 Cable block diagram for panel to panel**

- #1, Equipment tag number

  #1-1 Equipment supplied VENDOR:

  Tag numbers for field instrument, motors, panel, junction box and others shall be assigned as per project numbering procedure.

  #1-2 Equipment supplied CONTRACTOR:

  When tag number is not available, service (function) description shall be specified instead of tag number until it is determined by CONTRACTOR.

  Ex)

  Customer DCS panel

  Customer FGS panel

  Customer DCS junction box

  Customer UPS

  Customer Emergency power system

- #2, Explosion protection type:

  Explosion protection type shall be distinguished by employing following;

Input	Detail information
EXi	Ex i certified, (Intrinsic safety signal)
EXd	Ex d certified
EXp	Ex p certified
EXe	Ex e certified
EXm	Ex m certified
EXo	Ex o certified
EXq	Ex q certified
EXn	Ex n certified
NON EX	Non-certified.

- #3, IO tag or description:

  "Instrument IO tag number" or "load / signal description for electric system" shall be specified. Instrument IO tag number shall comply with project instrument IO tag numbering. For electric signal or load, VENDOR shall state the function description with relevant equipment tag number when IO tag number is not applicable. For instance, when cable is assigned for common fault signal for motor tag: PM001, **"Common fault, PM001"** shall be specified.

- #4, IO type:

  IO type shall be only selected from below table. Others are not acceptable. In addition, VENDOR shall ensure that the selection is in line with IO list and load list.

Input	Detail information
AI	Analogue input
AO	Analogue output
DI	Digital input
DO	Digital output (including 24V DC solenoid valve signal)
PWR	Power supply
COM	Serial communication / network
EAR	Earthing

- #5, IS/NIS:

  Intrinsic safety signal and non- Intrinsic safety signal shall be distinguished by employing following;

Input	Detail information
IS	Intrinsic safety signal
NIS	NON-Intrinsic safety signal

- #6, System:

  System shall be selected from below tables

  #6-1, for instrumentation

Input	Detail information
DCS	Customer distributed control system
ESD	Customer emergency shutdown system
SDS	Customer shutdown system
FGS	Customer fire gas system
DCS/UCP	Package distributed control system
ESD/UCP	Package emergency shutdown system
SDS/UCP	Package shutdown system
FGS/UCP	Package fire gas system
IE	Instrument earthing system
ISE	Intrinsic safety instrument earthing system

#6-2, for electric system

Input	Detail information
UPS	Customer uninterruptable power system
EMR	Customer emergency power system
ESS	Customer essential power system
NOR	Customer normal power system

Elec/UCP	Package electric control system
SE	Safety Earthing system

- #7, Current (A):

  Current value for circuit shall be indicated. Ex) 1A, 5A…….

- #8, Voltage:

  Voltage rate for circuit shall be shown. Ex) 24VDC, 230VAC, 6.9KVAC

- #9, Wire Qty

  Wire quantity requirement for signal and power transmission shall be specified. It shall contain shield (screen) and earthing wire requirement as well if applicable.

  Ex)

Input	Detail information
1C	1 core
2C	2 cores
2C+S	2 cores and 1 shield
3C+S	3 cores and 1 shield
2C+E	2 cores and 1 earth
3C+E	3 cores and 1 earth

- #10, Cable entry:

  Cable entry size of each cable shall be provided based on project cable gland selection table. The project cable gland selection table will be provided to VENDOR during detailed engineering stage. VENDOR shall select cable entry size from this table based on determined yard cable data. If “cable gland selection table” is not available at required stage, VENDOR can specify as “By CONTRACTOR”.

  In addition, VENDOR shall ensure cable entry size for field instrument and Ex d certified box is suitable for selected yard cable. If the standard entry size of equipment can’t accommodate yard cable/gland, VENDOR shall supply the proper reducer/adapter in order to meet yard cable / gland size.

  Figure 2 illuminates the sample of project cable and cable gland selection table.

Application	Cable Type	Cable Specification		Conductor Quantity	Cable	Remark	Cable Gland Selection (DONGA-BESTECH)
		Item	Specification		Code		
OGT	Flame Retardent Non-IS (Unarmour)	Outer Sheath	: Halogen free,Thermoplastic Compound,SHF1	1 Pair	**INUP01A**	TI(C)	SS2K - **20b**
		Overall Screen	: Aluminium / Polyester tape	2 Pair	INUP02A	TI(C)	SS2K - 20c
		Overall Earth Drain	: Tinned copper,according to IEC 60092-376	4 Pair	INUP04A		SS2K - 25
		Conductor Insulation	: XLPE	6 Pair	INUP06A		SS2K - 32a
		Conductor	: Stranded plain annealed copper,(class 2)	8 Pair	INUP08A	TI(C)	SS2K - 32a
			according to IEC 228	12 Pair	INUP12A	TI(C)	SS2K - 32b
		Insulation Grades	: 250V	18 Pair	INUP18A	TI(C)	SS2K - 40
		Outer Sheath Color	: Gray	24 Pair	INUP24A	TI(C)	SS2K - 50
		Conductor Insulator Color	: Pair Cable (White / Black)				
			Triad Cable (White/Black/Red)	1 Triad	INUT01A	TI(C)	SS2K - 20b
				2 Triad	INUT02A	TI(C)	SS2K - 25
				4 Triad	INUT04A	TI(C)	SS2K - 32a(S)
				6 Triad	INUT06A	TI(C)	SS2K - 32a
				8 Triad	INUT08A	TI(C)	SS2K - 32b
				12 Triad	INUT12A	TI(C)	SS2K - 40
				18 Triad	INUT18A	TI(C)	SS2K - 50
				24 Triad	INUT24A	TI(C)	SS2K - 50

Cable code

Entry size: M20

Figure 2 Project Cable / Cable gland selection table

- #11, Cable tag number:

  Cable tag number shall be specified according to project cable numbering.

- #12, Tray System:

  Cable tray system shall be indicated as per project cable tray segregation table.

Input	Detail information
HT	High voltage cable tray
LT	Power/UPS power/Electric Control cable tray
IT	Intrinsic safety cable tray (for instrumentation)

NT	Non Intrinsic safety cable tray (for instrumentation)
PA	PAGA A system cable tray
PB	PAGA B system cable tray
CT	Communication / network cable tray

- #13, Cable type:

  Cable type shall be indicated based on project cable type code. The project cable selection table will be provided to VENDOR during detailed engineering stage. So, all yard cable code shall be selected from this table by VENDOR. If “project cable selection table” is not available at required stage, VENDOR can specify as “By CONTRACTOR”.

  Refer to Appendix_ A for the sample of case 1.

Case 2: Junction box to control panel.

It shall be submitted via project template of “junction box to control panel”. VENDOR cables which are installed by VENDOR shall not be included.

When junction box supplied by CONTRACTOR is connected to control panel supplied VENDOR, the responsibility of this drawing is still under VENDOR.

For instance, when field instruments and junction boxes for CCC (compressor Controls Corporation) are supplied by CONTRACTOR, COMPRESSOR VENDOR shall provide cable block diagram between junction box and unit control panel based on project junction box segregation rules and CONTRACTOR will confirm and comment it.

Refer to Appendix_ B for the sample of case 2.

Case 3: Field devices to panel.

Template for “field devices to panel” shall be used. When more than two cables are required for one field devices, VENDOR should segregate the cables as per project segregation requirement.

Even though the supply scope of VENDOR is limited to field equipment such as motors or field instrument, VENDOR still has responsibility to submit “tie in cable block diagram” in order to determine yard cable type and cable entry sizes.

VENDOR shall ensure not to indicate VENDOR cables.

Refer to Appendix_ C for the sample of case 3.

Case 4: Field devices to Junction box

It shall be generated via template for “field devices to junction box”. Although junction box is not supplied by VENDOR, VENDOR shall submit this drawing for his ship loose devices (instruments) which connects yard cable. Field devices (instrument) grouping for each junction box shall be conducted as per project junction box segregation requirement. Junction box tag number supplied by CONTRACTOR will be informed during vendor data review stage.

VENDOR shall exclude VENDOR cables from this drawing.

Refer to Appendix_ D for the sample of case 4.

4.2 Tie in Cable schedule

Tie in cable schedule shall be submitted after receiving 1st comment on tie in cable block diagram from CONTRACTOR. It is VENDOR's responsibility to incorporate all comment /data on tie in cable block diagram to this document and to exclude VENDOR cables installed by VENDOR from this DWG. It is important to specify the data of terminal block assigned for yard cable by employing the project cable schedule template

4.3 Tie in Termination Drawing

Tie in termination DWG can be submitted via VENDOR's format. However, it is crucial requirement to include following information for yard cable as minimum.

A. Cable tag numbering

B. Cable type

C. IO tag / type for instrument (Note 1)

D. Power load / Power signal description for Electric (Note 1)

E. From / To equipment tag number

F. Terminal block strip number

G. Terminal block number

H. Yard cable screen/shield termination

I. Power rating/consumption

J. Diode / resistance for line monitoring if applicable

Note 1: For instrumentation, it shall indicate the "IO tag number" for each termination point. On the other hands, "power load/power signal description" will replace with "IO tag number" in case of electric termination.

Figure 3 & Figure 4 show the guidance of above minimum data requirement. In addition, termination detail for VENDOR cables shall not be shown on this drawing.

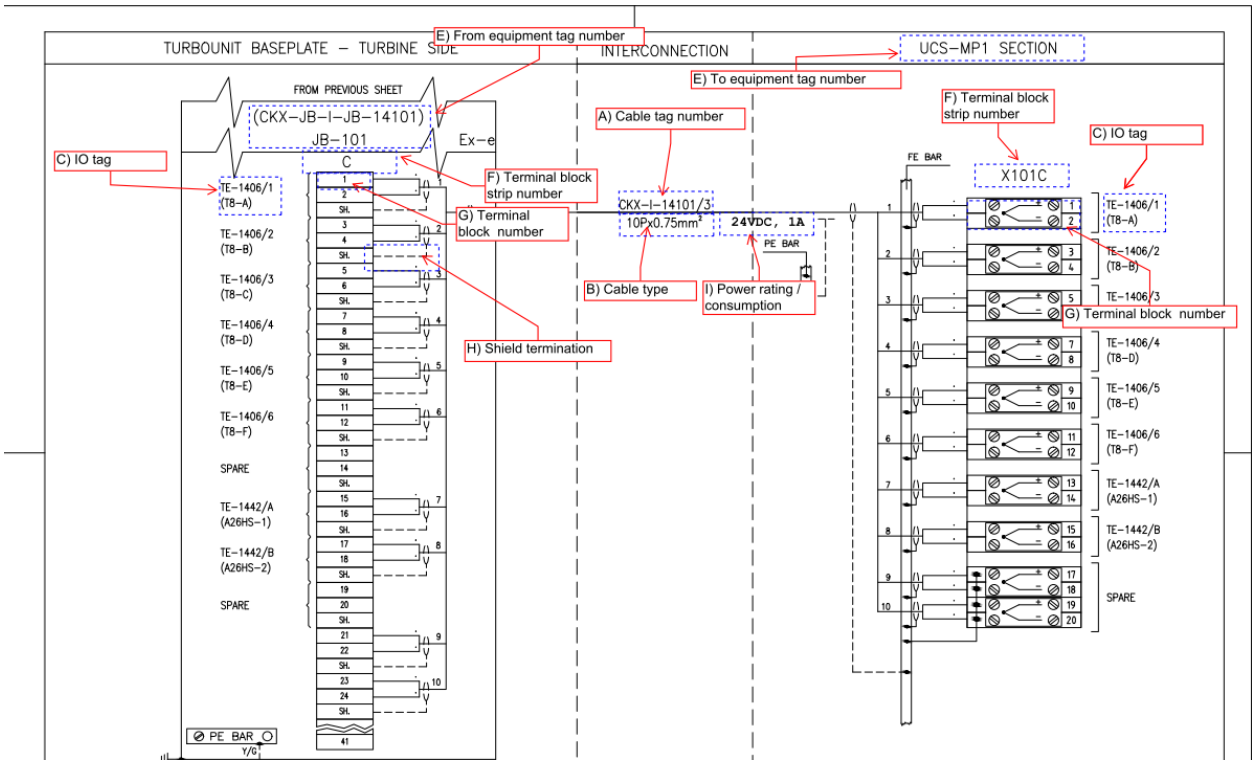


Figure 3 Tie in termination For INSTRUMENT

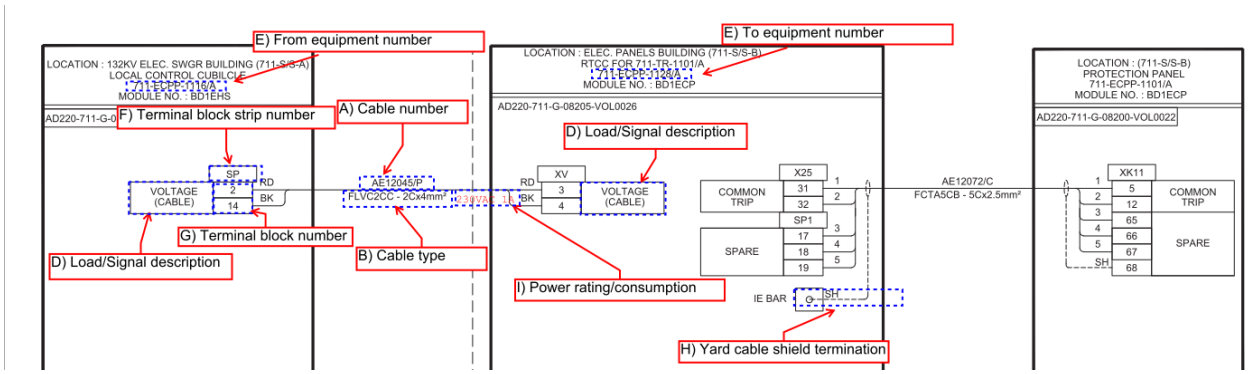


Figure 4 Tie in termination for ELETRIC

In addition, if there is any communication interface between customer equipment and VENDOR's cabinet, VENDOR shall include the connection DWG for network cables such as FO cables and Cat 6. Connection DWG for network cable shall clearly indicate component/switch tag number, cable tag number and from/to equipment tag number as per Figure 5. If there is more than one port for any network devices such as switch and firewall, VENDOR shall ensure to indicate port number that interface network cable to be connected.

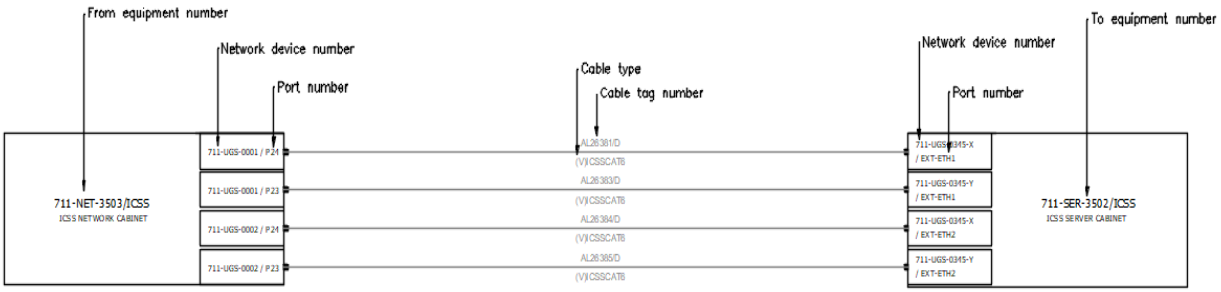


Figure 5 Network Cable Connection Drawing

4.4 Tie in E&I layout

This document is key drawing to determine yard cable/tubing routing. Following information shall be shown as minimum.

A. The location of E&I equipment.

Note: E&I equipment tag number shall be matched with the data on tie in cable block diagram

B. Cable tray between skid edge tie in location and E&I equipment. (with tray dimensional data)

Note: Only cable tray on which laid yard cable need to be shown.

C. Skid edge tie in location / points (for cables/tubing)

D. Yard cable list with assumed “in skid length” / tubing list for interface

E. Cable tray spare space data after installing yard cables.

F. Cable tray segregation

It is recommended that separated sheet for cable tray arrangement including E&I equipment location, cable routing with cable list & spare space data and tubing routing to be submitted for good visibility.

CONTRACTOR will inform the preferred skid edge tie in point (for cables / tubing) to VENDOR during detailed engineering. VENDOR shall arrange cable/tubing tray up to points that CONTRACTOR informs. If the location provided by CONTRACTOR is not feasible, alternative location shall be agreed with both parties during E&I tie in review meeting (On line/offline).

As per Figure 6~Figure 13, tie in E&I layout shall comply following minimum requirement/instruction.

- Instruction#1: Tie-in location (instruction #1) shall be indicated with X, Y, Z coordination and skid cable tray shall be installed up to skid edge.
- Instruction #2: Cable tray size/length/system to be shown. VENDOR shall ensure to show cable tray cover as well if applicable.

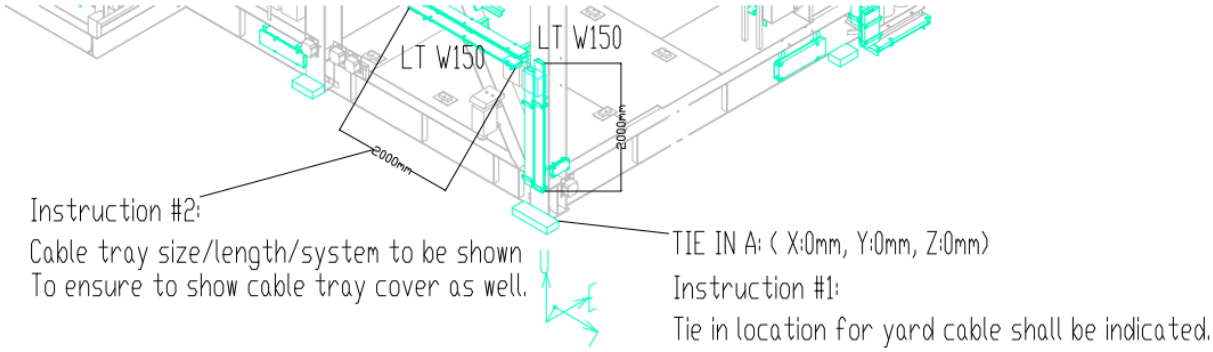


Figure 6 E&I Tie in layout instruction #1~2

- Instruction #3: Cable tray shall be shown up to destination with equipment tag number as per Figure 7.

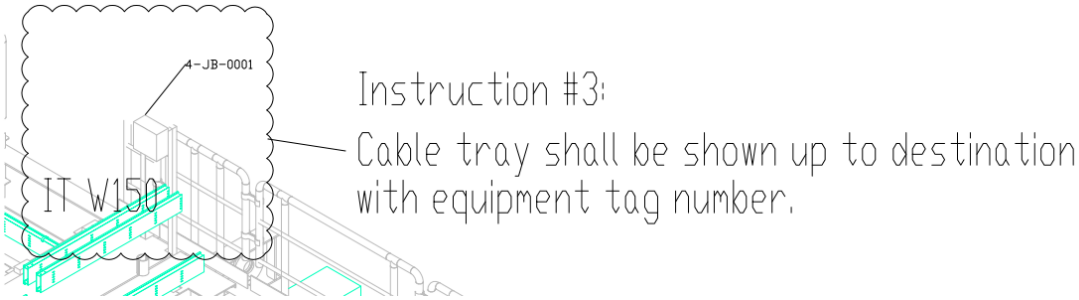


Figure 7 E&I Tie in layout instruction #3

- Instruction #4: Tray shall be continually connected by using tray fitting such as splice unit, elbow or adjustable fitting. Otherwise, it shall be approved via TQ.

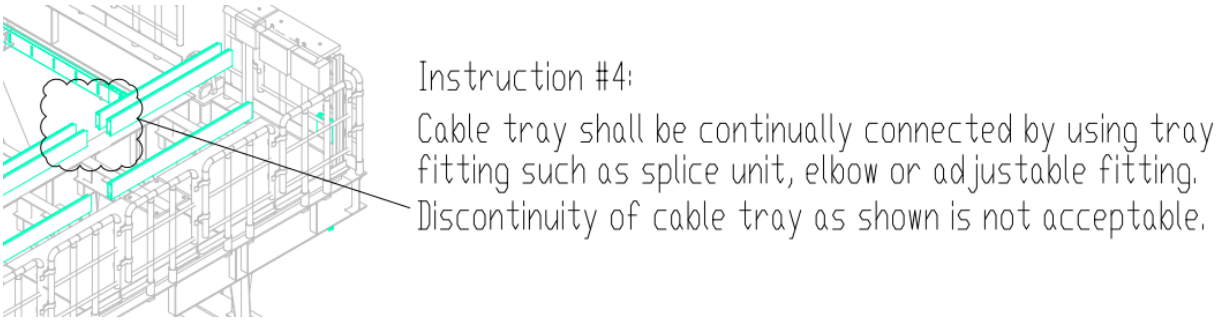


Figure 8 E&I Tie in layout instruction #4

- Instruction#5: Tag number for E&I equipment which need to be connected with yard cables shall be indicated. In addition, all tag number shall be in line with those in tie in cable block diagram.

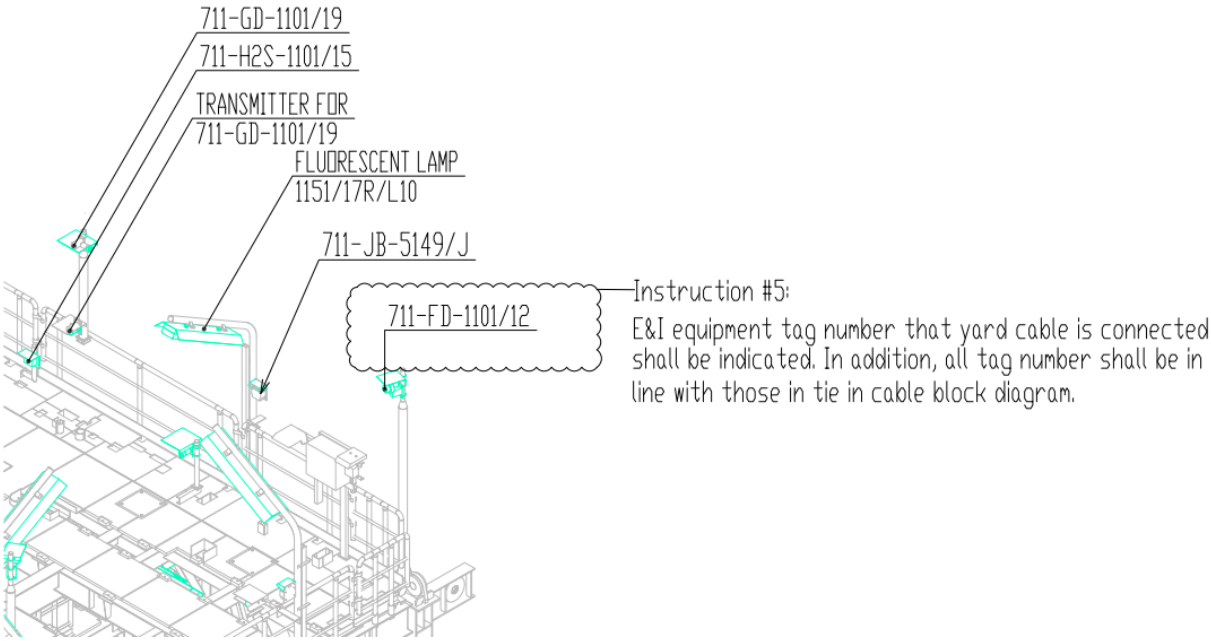


Figure 9 E&I Tie in layout instruction #5

- Instruction#6: Special attention shall be given to provide the enough clearance for yard cable bending radius at cable entry of E&I devices. The space for yard cable bending and cable entry of each devices shall be specified as per Figure 10. The required clearance shall be calculated based on the largest cable which is terminated to each equipment.

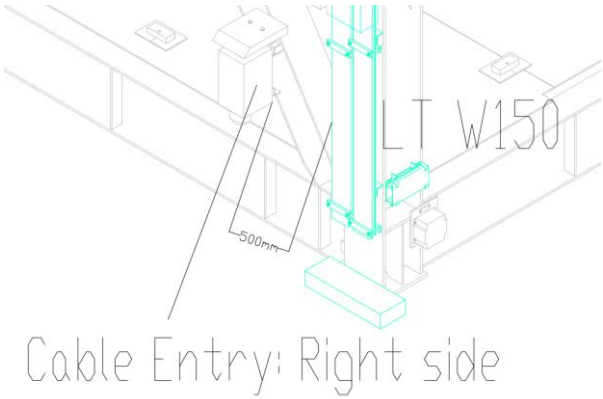
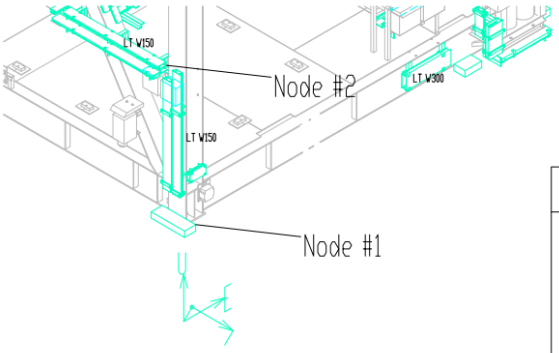


Figure 10 Cable Bending Radius Check at Cable Entry

- Instruction#7: Yard cable list and spare space data for each tray node shall be provided. It is ensured that every tie in point and MCT (multi cable transit) shall be included in one of cable tray nodes.



Node No.	Cable list	Tray spare %
#1	NT-JB-0001 NT-MT-0002 NT-JB-0002 NT-MT-0009	45%
#2	NT-JB-0001 NT-MT-0002 NT-JB-0002	70%

Figure 11 Cable list and Spare space data for each tray

- Instruction#8: Cable tray segregation shall be provided as per project requirement. The distance between different tray system shall be indicated and if any deviation due to physical limitation, it shall be approved via official technical queries. The distance data on this drawing will not relieve VENDOR's responsibility of segregation requirement. Figure 12 illuminates the distance between NT (Non IS instrument cable tray) and LT (LV power cable tray).

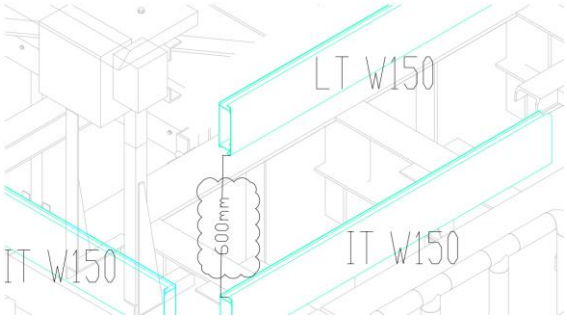


Figure 12 Cable tray segregation

- Instruction#9: If tubing interface such as fusible plug system is applicable to the package, tubing and support shall be arranged up to skid edge tie in point. Only for vessels and tanks package, it is allowed to provide only support for yard fusible line/tubing on vessel skid without installing tubing. In addition, if passive fire protection/insulation is employed on vessel skid, the support of fusible tubing shall be extended sufficiently to avoid to be buried inside passive fire protection/insulation of vessel.

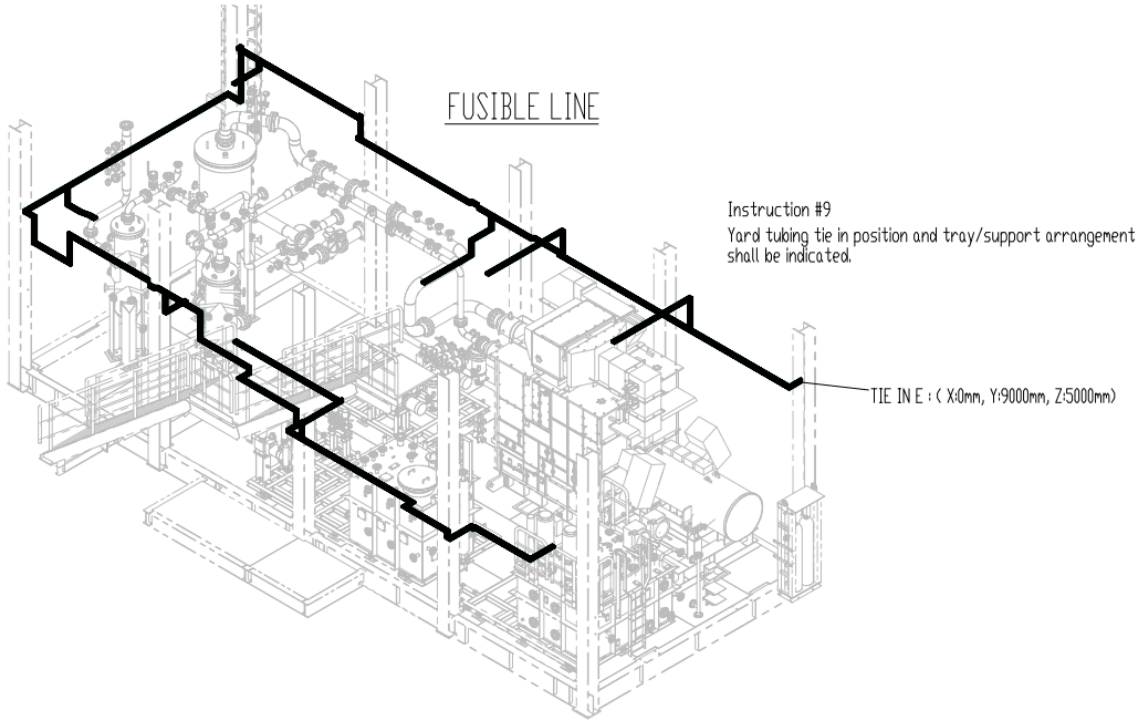


Figure 13 Tubing Tie in (Ex. Fusible line)

In addition, VENDOR shall provide bolted type cable tray support nearby tie in point according to COMPANY acceptance during detailed engineering in order to adjust tie in location in case of mismatching. If it is approved, bolted type support can be only applied to the approved area.

4.5 Tie in Panel/Cabinet layout

It is critical document for the control room arrangement. It shall show the information of panel foot print with fixing hole detail, door opening, gland plate hole arrangement and termination points for each yard cables.

Panel foot print shall indicate clearly fixing holes size as well as position of holes as per Figure 14. On top of that, VENDOR shall specify recommended fixing bolt size and toque value.

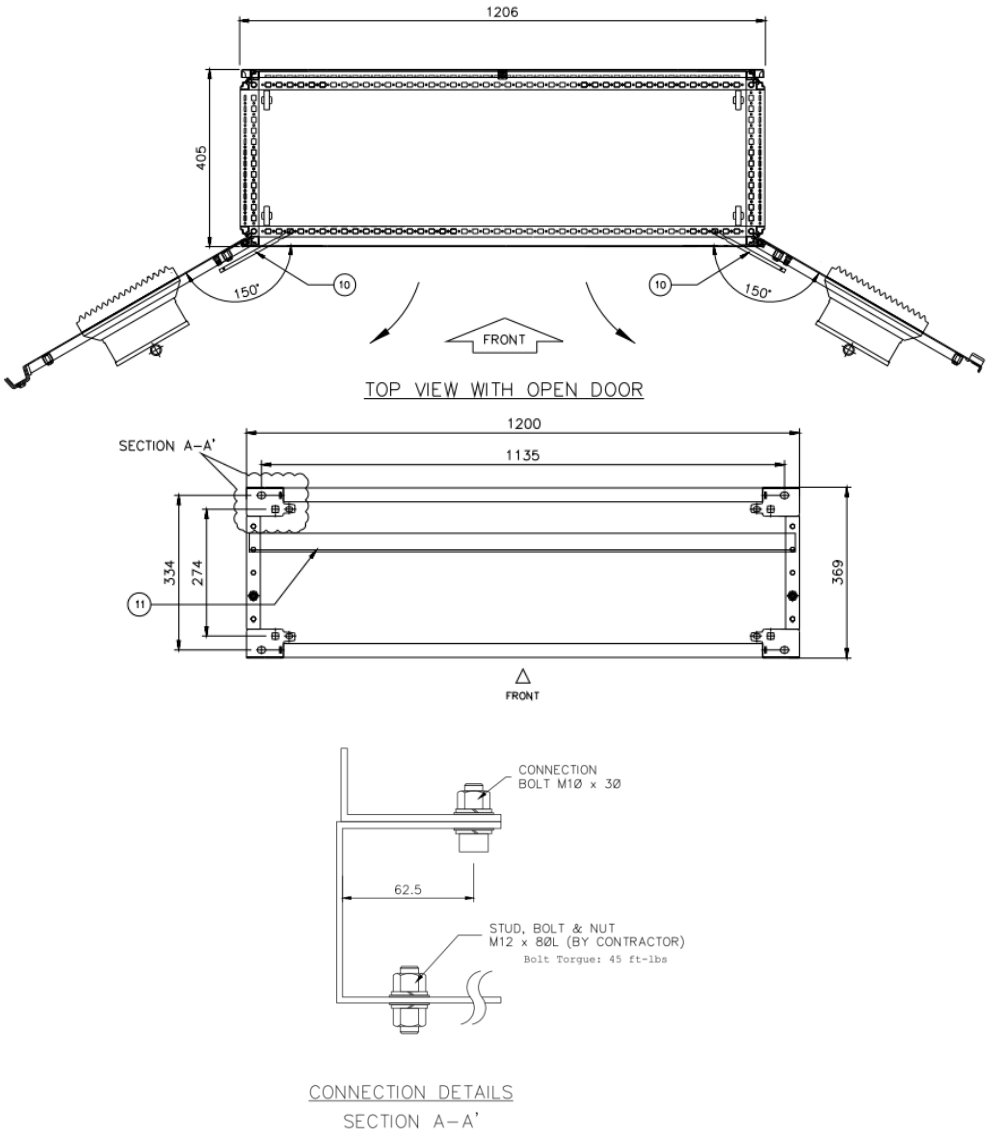
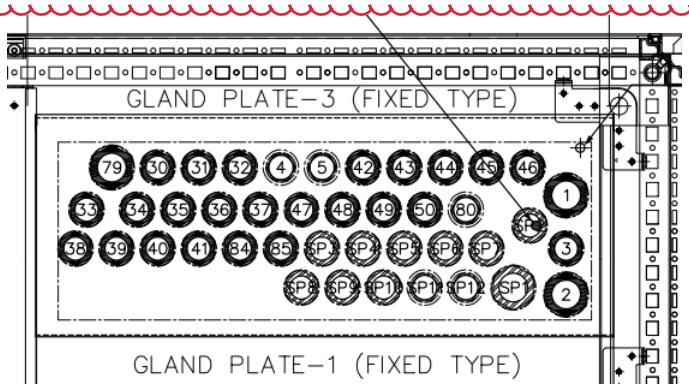


Figure 14 Door opening and foot print

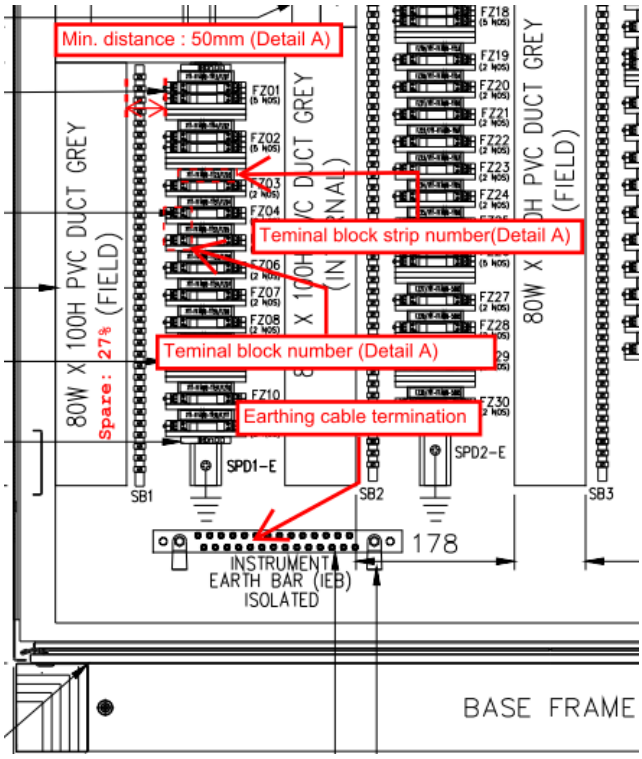
Gland plate hole arrangement is required to check whether proposed gland plate size is enough to accommodate all tie in cables. If VESDA (very early smoke detection apparatus) is applicable, entry hole for VESDA pipe interface shall be indicated on gland plate hole arrangement drawing. Furthermore, VESDA tubing within cabinet shall be installed up to tie in location by VENDOR. Figure 15 illuminates gland plate hole arrangement as well as relevant cable list.



HOLE No	GLAND SIZE	HOLE SIZE	Cable No	TB No
1	M25	ϕ26	AE05650/P	Power cable
2	M25	ϕ26	AE05671/P	Power cable
3	M20D	ϕ21	AE05952/P	Power cable
4	M20C	ϕ21	AE26580/E/1	Earth cable
5	M20C	ϕ21	AE26580/E/2	Earth cable
6	M20C	ϕ21	AE26580/E/3	Earth cable
7	M20C	ϕ21	AE26580/E/4	Earth cable

Figure 15 Gland plate holes arrangement

Termination points for each tie in cable shall be indicated on this drawing as per Figure 16. All terminal block strip number shall be in line with data on tie in termination DWG. In addition, VENDOR shall make sure earth bar and shield clamp shall be arranged for yard grounding cable interface.



--Detail A--

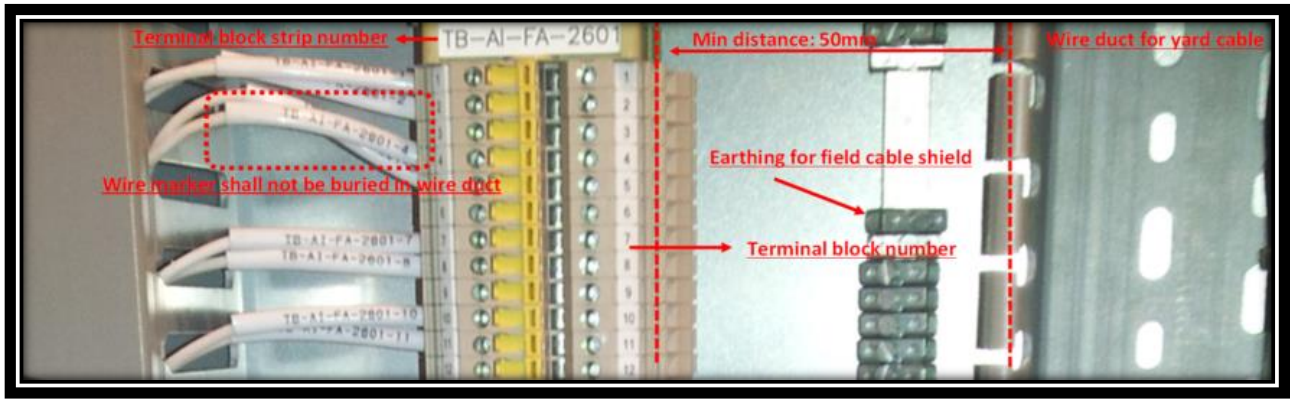


Figure 16 Tie in cable termination information in panel

Wire duct size for yard cable shall be sufficient to accommodate all yard cables considering spare

requirement. The size, spare rate and color of wire duct shall be indicated. Duct color for intrinsic safety signal shall be blue as per IEC requirement. For wire duct for others, color shall be in line with project requirement.

The distance between wire duct and termination components such as terminal block shall be indicated on the drawing. It is not acceptable when any wire marker is buried in the wire duct due to not enough space. So, the distance shall be minimum 50mm to keep the visibility of wire marker.

4.6 Tie in E&I installation detail for loosed devices

Tie in E&I installation detail shall indicate the mounting method for all ship loose E&I equipment which is installed by CONTRACTOR at yard including required bill of material. Figure 17 is the example of ship loose instrument which need to be installed by CONTRACTOR. It is crucial to indicate the mounting holes size on mounting bracket/plate as well as recommended mounting bolt size with torque value.

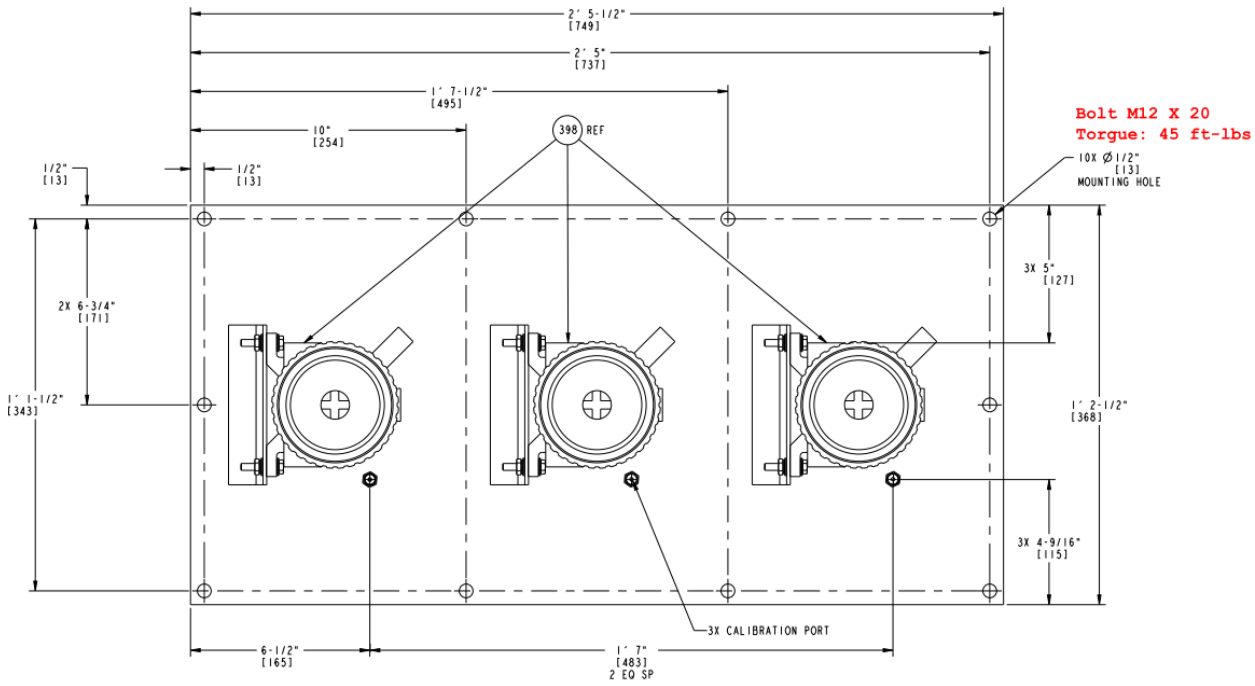
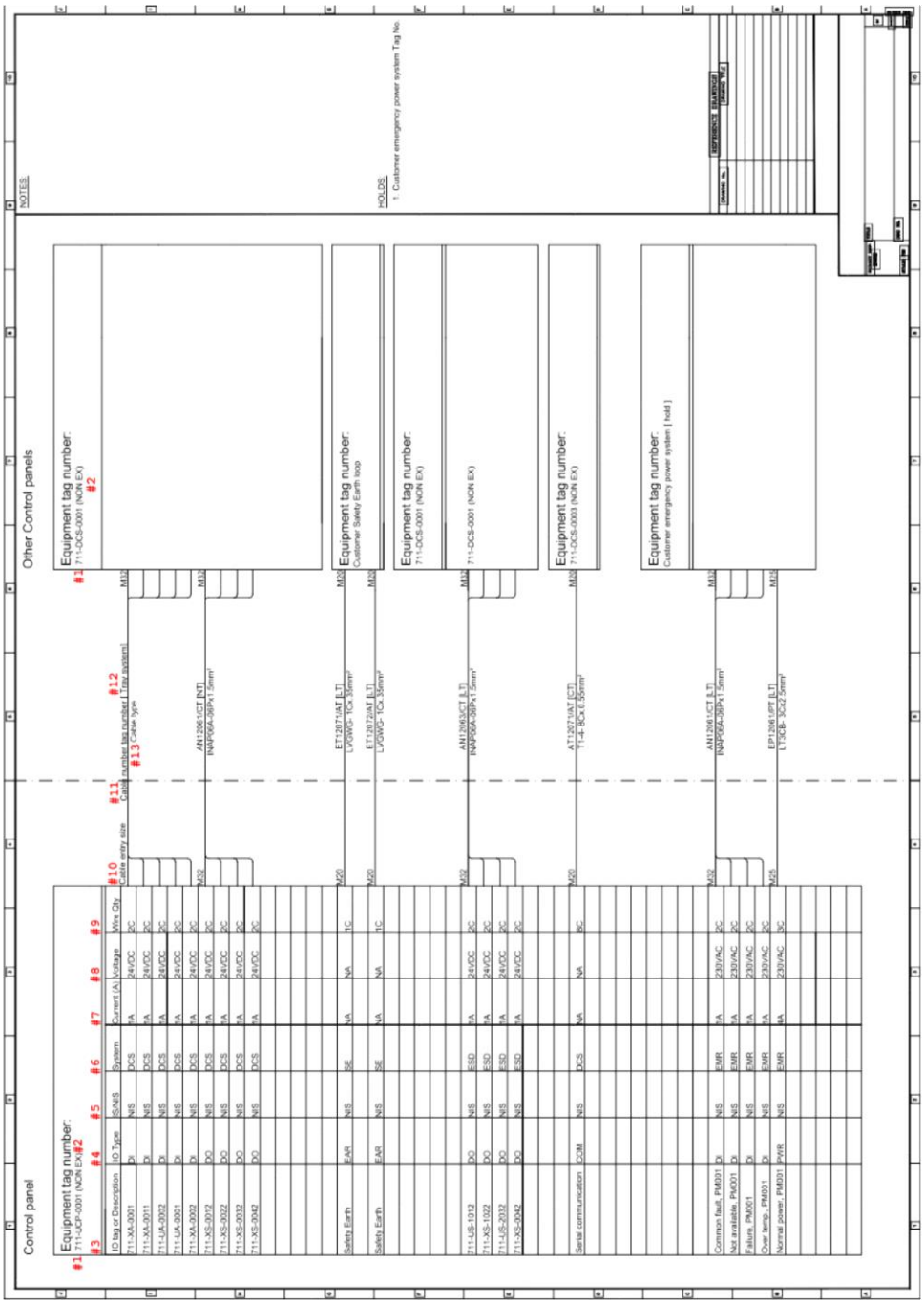
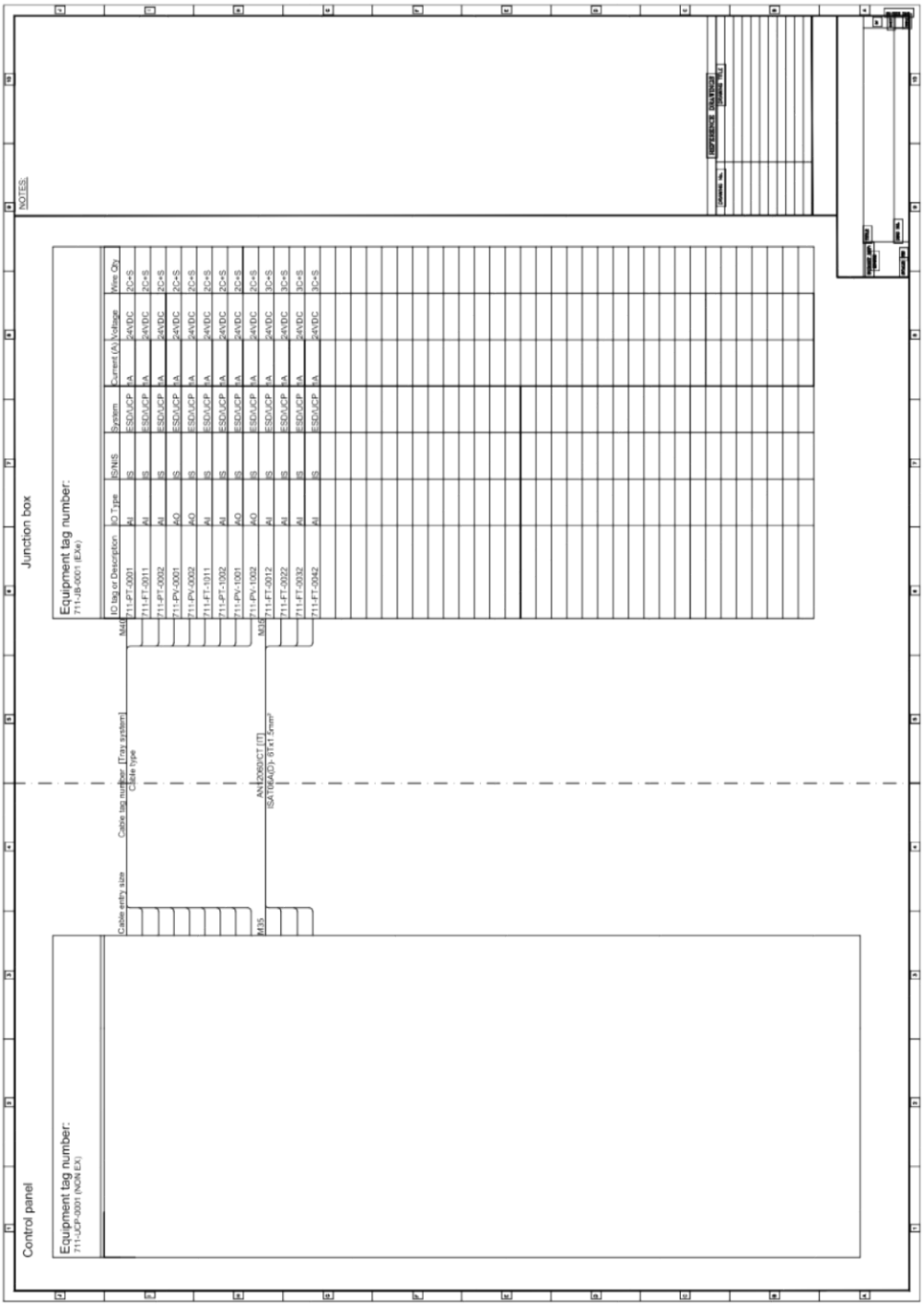


Figure 17 Tie in E&I installation detail

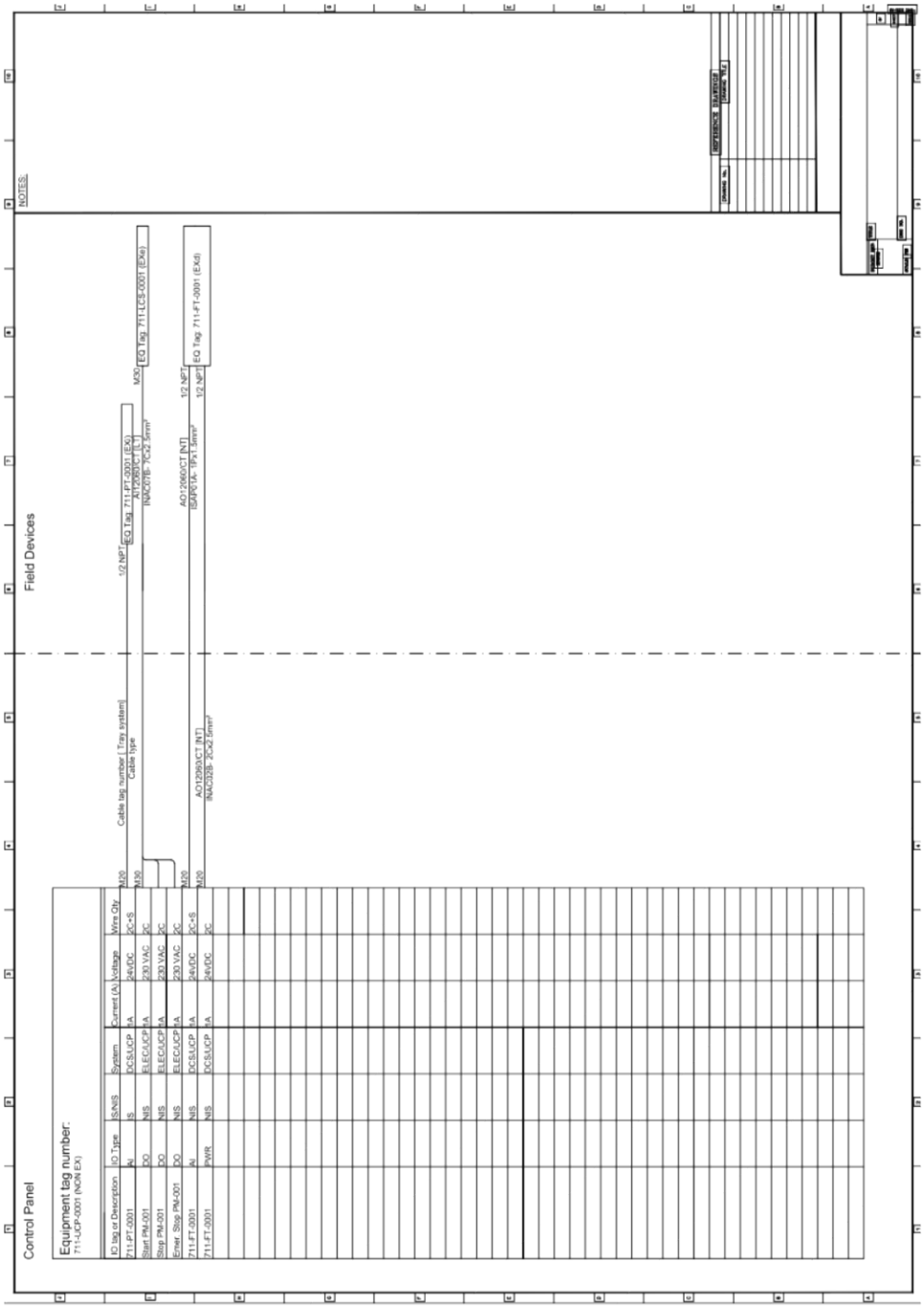
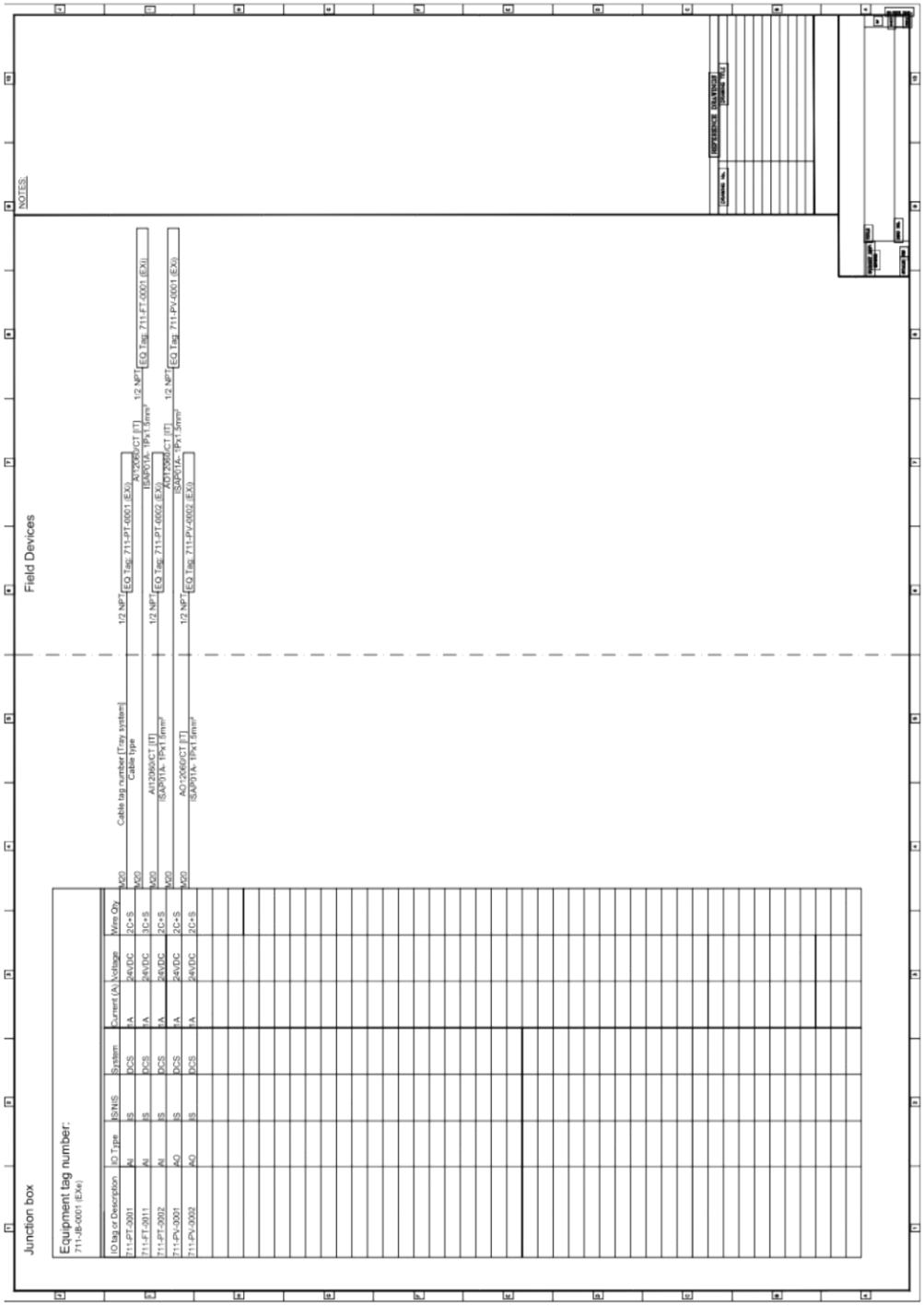
APPENDIX- A. Cable block diagram for panel to panel



APPENDIX- B. Cable block diagram for Junction box to panel



APPENDIX- C. Cable block diagram for field devices to panel



APPENDIX- D Cable block diagram for field devices to junction box